# SENIOR SOFTWARE / AI ENGINEER

## ROMAN GIRIN, Ph.D.

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## PROFESSIONAL SUMMARY

Senior Software & Applied AI Engineer with 20+ years of experience specializing in high-load backend systems, enterprise cloud migrations, and production-grade AI infrastructure. Combines an advanced academic foundation (Ph.D. in Applied Neural Networks) with decades of hands-on mastery across the .NET ecosystem, microservices, and applied machine learning (Python, PyTorch, LLMs).

## KEY SKILLS & TECH STACK

• **Programming Languages:** C#, Python, SQL (T-SQL, PL/pgSQL), C++
• **Platforms & Frameworks:** .NET 8 / Core / Framework, ASP.NET, Entity Framework Core, Django REST Framework
• **Software Architecture & Patterns:** Monolith-to-Microservices, Domain-Driven Design (DDD), Event-Driven Architecture (EDA), CQRS, Saga, Outbox, RESTful API, HATEOAS, SOLID, GoF
• **AI Orchestration & Backend (.NET):** MS Semantic Kernel, LLMs API, RAG, pgvector (Vector Search), Lunary (Production Performance & Observability)
• **Applied ML & MLOps Frameworks (Python):** PyTorch, Hugging Face Transformers, LoRA Fine-Tuning, OpenCV, TorchVision, Scikit-Learn, EleutherAI
• **Message Brokers & Streaming:** Azure Event Grid, Azure Queues, Kafka, RabbitMQ
• **Databases:** MS SQL, PostgreSQL, noSQL Azure Table Storage
• **Cloud, DevOps & Observability:** Docker, Azure Kubernetes Service (AKS), Azure Container Apps, Azure DevOps (CI/CD pipelines)
• **Testing & Engineering Culture:** xUnit, NSubstitute, Test-Driven Development (TDD), RAG Semantic Search Testing, LLM Function Invocation Testing
• **AI Productive Tools:** Claude Code, LM Studio

## PROFESSIONAL EXPERIENCE

### River Logic, Inc. | Seattle, USA (Remote)

*Senior Modeling Software .NET Developer / AI Engineer* | *March 2014 - October 2026*

**AI & LLM ENGINEERING**

• **AI Orchestration into .NET Ecosystem:** Architected and deployed production enterprise models orchestration and agentic workflow with RAG leveraging vector embeddings (pgvector). Built an MS Semantic Kernel plugin to integrate core software via ASP.NET 8 REST HATEOAS API, enabling tooling integration into LLM assistant. AI assistant monitoring in production via Lunary

• **Python-Driven LLM R&D:** Conducted target R&D and executed open-source LLM fine-tuning via LoRA using the Transformers library (Python, PyTorch) to customize, optimize, and align open-source models for domain-specific enterprise workflows. Post-fine-tuning model evaluation via EleutherAI.

• **R&D in Mathematical Optimization (Graph ML):** Designed and built a Proof of Concept (PoC) lightweight model using Graph Neural Networks (GCNN) to estimate LP (Linear Programming) task feasibility pre-solve with 92% accuracy. Mapped LP matrix representations to bipartite graphs using PyTorch Geometric (PyG).
• **Team Upskilling:** Introduced agentic AI workflows (Claude Code) across the engineering team, raising development velocity.

**SOFTWARE ARCHITECTURE & .NET DEVELOPMENT**

• **Architecture Modernization:** Spearheaded the architectural transition from a legacy 2-tier to 3-tier architecture. Migrated Business Logic Layer (BLL) from database-tier (MSSQL T-SQL) into the .NET application layer (C#), improving system maintainability resulting in a 30% reduction in support ticket volume. Implemented compatibility with PostgreSQL.

• **Windows to Linux Migration:** Spearheaded the core codebase migration from .NET Framework to cross-platform .NET 8, implementing a modern asynchronous programming model (async/await) that slashed system resource utilization by 15%.

• **Microservices in Cloud:** Decomposed an on-premise monolith into high-load microservices deployed on Azure Container Apps and Azure Kubernetes Service (AKS). Designed inter-service communication using REST and asynchronous messaging EDA (Azure Event Grid / Queues), ensuring distributed data consistency via Saga, CQRS, and Outbox patterns.

• **Native Interoperability:** Developed critical components in C++ to seamlessly integrate CLR-managed and native code blocks.

• **Engineering Culture:** Practiced DDD, SOLID, GoF. Established testing standards by implementing comprehensive unit and integration testing (xUnit, NSubstitute) and Test-Driven Development (TDD). Managed build and deployment workflows via Azure DevOps pipelines.

## Ecosmart LLC | Samara, Russia

Senior Computer Vision Engineer *(Part-Time)* | *September 2019 – October 2026*

**ANPR-SYSTEM DEVELOPMENT**

• **System Design:** Designed and implemented an Automatic Number Plate Recognition (ANPR) system driven by custom deep networks (MTCNN, STR-net, OCR).
• **CNN Architecture Optimization:** Conducted deep structural analysis on MTCNN. Re-engineered the detection pipeline using P-Net and O-Net while eliminating R-Net completely, reducing GPU compute load with zero degradation in accuracy.
• **Real-time processing**: Implemented the main video frame processing cycle in C++ to achieve up to 11–13.5 ms (~75–90 fps) processing speed (about 1.5× faster than Python implementation).
• **Object tracking**: Implemented optimized to car tracking application module based on Deep-\BoT-\Strong-SORT (Simple Online Real-time Tracking). Performed fine-tuning of OSNet (Omni-

Scale NET) to car visual appearance embedding extraction achieving re-ID performance at mAP 72%, Rank-1 94%.
• **Data Management:** Customized the LabelImg tool to support proprietary company metadata formats. Led a team of data annotators and built the dataset validation pipeline. Configured an on-premises pipeline for data versioning and experiment tracking via MLflow and DVC.
• **Business Impact:** The system has been successfully deployed at a number of facilities, leveraging automatic license plate recognition (ALPR) to provide advance notification of expected road trains approaching the unloading warehouse, thereby completely eliminating delays. It also ensures the optimization and tracking of the industrial enterprise's corporate fleet, alongside automated access control gate management at various residential and industrial sites.

**CCTV-BASED COMPLIANCE AUDIT SYSTEM DEVELOPMENT**

• **Detection System Development:** Built and trained an object detection model based on the SSD (Single Shot MultiBox Detector) architecture to track specialized commercial cleaning vehicles in real-time video streams.
• **Data Engineering:** Automated data collection, filtering, and image preprocessing routines optimized for non-standard and changing indoor lighting conditions.
• **Business Impact:** Deployed the monitoring system across "Lastochka" express train maintenance depots, effectively eliminating cleaning schedule violations and entirely preventing compliance penalties from regulatory bodies.

## Samara State Technical University | Samara, Russia

*Computer Vision Engineer (Part-Time / Fellowship)* | *September 2017 – September 2019*

• **System Design:** Engineered a machine vision diagnostic system utilizing Deep Convolutional Neural Networks (CNNs) to automate the detection of broken railway fasteners and bindings using Python, PyTorch, OpenCV, TorchVision, and CUDA/cuDNN acceleration.
• **Dataset & MLOps:** Gathered imaging dataset for CNN training, successfully managed severe data class imbalances by implementing synthetic generation and augmentation techniques via `imbalanced-learn` (SMOTE).
• **Optimization & Edge Deployment:** Converted models into TensorRT and ONNX formats using FP16/INT8 quantization to optimize models for real-time inference on Edge computing platforms (NVIDIA Jetson Nano and AGX Xavier).
• **Integration & Async Backend:** Tied the imaging dataflow into enterprise infrastructure via a custom REST HATEOAS API (Django REST Framework) and asynchronous messaging streams (Kafka/RabbitMQ).
• **Camera Calibration:** Experienced in configuring Ethernet cameras for CV systems, specifically working with scientific Baumer cameras and their neoAPI.
• **Business Impact:** Enabling Jetson-powered inference inside mobile laboratory trains JSC RPC "InfoTrans" allowed defect discovery at full transit speeds, boosting defect detection rates by 60%.

## VDcom | Samara, Russia

*Senior Software Specialist (Promoted for Outstanding Performance)* | *May 2010 – March 2014*

• **Outstaffing at Sberbank CIB:** Developed backend functionality on the Diasoft platform for the investment banking business.
• **High-load Databases:** Designed, developed, and optimized complex stored procedures, ER-models and data-tier components for high-load transaction systems across Sybase ASE and MSSQL database engines.
• **Technical Estimation:** Conducted technical analysis and pre-sale engineering evaluations for resource and cost estimation.

## ACS – VolgaEVMcomplex | Samara, Russia

*Project Manager / .NET Developer* | **February 2006 – May 2010**

• **Systems Analysis & Design:** Partnered with clients to formalize requirements into detailed Technical Specification Documents (TSDs) for large-scale business process automation.
• **Full-Stack C# Engineering:** Managed end-to-end Enterprise Content Management (ECM) implementations while actively writing custom C# extensions and core integration modules.

## EDUCATION & RESEARCH

**Samara State Technical University (SamSTU)** | Samara, Russia
*Ph.D. (K.T.H.) in Technical Sciences* | *Specialization: Applied Neural Networks* | **Graduated 2019**

• **Thesis:** “Intelligent informational measuring system for thermography-based diagnostic of technical objects based on artificial neural networks.”
• **Research Portfolio:** Published academic insights available at ResearchGate https://www.researchgate.net/profile/roman-girin/research .

**Samara State Technical University (SamSTU)** | Samara, Russia
*Engineer’s Degree (5-year Specialist Degree with Honors)* | **Graduated 2002**

## CERTIFICATIONS & PORTFOLIO

• **Languages:** English (Full Professional Proficiency / C1), Russian (Native)
• **Certifications:** Federal Financial Markets Service (FFMS) Qualification Certificate for Financial Market Specialists (2008)
• **Portfolio:** https://romangirin.github.io/
• **Pet projects:** Designed, implemented, and deployed into production a series of pet projects; detailed descriptions and demos are available upon request.